Project Progress Report

Your project progress report should include the following.

Proposals should include:

- An introduction to the problem
- Brief Overview of Modelling that has been used to tackle similar problems
- Model Formulation and Justification
- Preliminary Results
- Future work
- References

To further explain things I include an example below.

REF means that there are references cited in that location. (i.e., cite papers, books, news sites, etc)

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1. Introduction

[Make sure that any information that we need to know about your problem is listed here. For example, if I am writing a modelling proposal about something in biology I make sure that I introduce the biology in the introduction, the problems around understanding the biology of the problem, and why a model of the system is needed to further understand the process.]

Escherichia coli (E-coli), discovered by Theodor Escherich (REF1), is a gram negative bacterium that can live in the intestines of mammals. This bacterium is known for causing food poisoning in humans and has caused some serious outbreaks of disease including water contamination (i.e. Walkerton Ontario (REF2)) as it can survive for brief periods in fecal deposits that, when untreated in sewage treatment plants, will be re-introduced into the environment (REF1, REF2).

E-coli infections are sometimes treated with antibiotics (REF3). However, since there are many strains of E-coli, which vary widely, and since E-coli is gram negative, it is resistant to many antibiotics that have been proven effective in treating other bacterial infections (REF3).

The evolution of resistance to many antibiotics has led researchers to develop other control mechanisms for this pathogen. One control is in the form of phage therapy (i.e. viruses that target pathogenic bacteria). Currently, phage therapy is not approved for use in humans in Canada (REF4).

Another control is in the form of vaccination. In March 2006, it was reported that researchers have been successful in developing a vaccine that elicits an immune response to E-coli. This vaccine has been reported to be safe in children 2-5 years old and in adults (REF5). Currently, there is a clinical trial underway to verify its efficacy against E-coli (REF5).

If or when this vaccination is approved, a vaccination distribution strategy will be required so as to effectively administer the vaccine to control the pathogen and protect susceptible individuals in a population. We propose a new model of E-coli spread in a population, including antibiotic treatment, vaccination and the effects of pathogen viability in a water reservoir. Such a model will enable us to predict severities of E-coli outbreaks, depending on contamination levels of the water reservoir, contact

between individuals in the population, the use of antibiotics and the evolution of resistance to these drugs, and different vaccination control strategies. It is evident that this model, and it's predictions, may have an effect on the development of health control policies and procedure.

2. Literature Review

[Refer to previous models of e-coli, the use of antibotics to treat bacterial infection, the effects of antibiotic resistance]

The modeling literature around bacteria, antibiotics, phage therapy, and vaccines is a complex field that seeks to understand and optimize approaches to combating bacterial infections, especially in the face of rising antibiotic resistance. Below is a brief overview of the key areas, along with references for further reading:

Mathematical models have been used to describe bacterial growth and the evolution of antibiotic resistance. These models typically use differential equations to represent the dynamics of bacterial populations under the influence of antibiotics and how bacteria evolve mechanisms to resist them. Models often incorporate factors such as mutation rates, horizontal gene transfer, and selective pressure induced by antibiotics (REFs). Some models focus on the impact of dosing regimens, duration of therapy, and drug combinations on the emergence of resistance (REFs). Some studies apply optimal control theory to determine the best antibiotic treatment regimens that can minimize both bacterial growth and the risk of resistance development (REFs).
There is also modelling research on how antibiotic cycling (rotating antibiotics) or combination therapies can slow resistance (REFs).

Phage therapy has re-emerged as a promising alternative to antibiotics, particularly for multidrug-resistant bacterial infections. Mathematical models of phage therapy explore the dynamics of phage-bacteria interactions and how phages can be used to control bacterial populations. Models consider the rate of bacterial lysis, the replication rate of phages, and the potential for bacterial resistance to phages (REFs). Some research integrates phage therapy with antibiotics, showing that phages can help restore the effectiveness of antibiotics (REFs).

Vaccines are seen as a key strategy to reduce the burden of bacterial infections and antibiotic use. Vaccine modeling focuses on understanding how vaccination can reduce the transmission of bacterial infections and help limit antibiotic resistance. There are many models of vaccination programs (REFs – give a long list of some of them). Some models have been used to predict the potential impact of new vaccines on both public health and the emergence of antibiotic-resistant strains (REFs) including the effects of bacteria spread through environmental reservoirs (i.e., water) (REFs). We will build on modelling literature to explore effective vaccine rollout for a new vaccine against E-coli.

3. Model

[Introduce the model, justify the structure, include a table of parameters and variables]

We propose a model of E-coli spread, including susceptibles, infected, and recovered individuals in a population. It is assumed that (1) susceptibles individuals can be vaccinated against E-coli, (2) infected individuals can obtain antibiotic treatment and (3) recovered individuals will confer immunity to E-coli. We will model the evolution of drug resistance as the probability of mutation to full resistance in the infected individuals (thus there will be more than one infected class). It is assumed that susceptible individuals may be infected by transmission from infected individuals, or by exposure to contaminated water reservoirs. Shedding of the bacteria from infected humans into the infected reservoir will be

included in the model. A diagram of the model is shown in Figure 1. Parameter values for our model are listed in a table with references to other studies that measured these values. Model equations are given in Equation (1).

$$
\begin{aligned}
S' &= \lambda - dS - \beta_Y SY - \beta_X SX - \beta_e SE_Y - \beta_e SE_X - \rho S \\
V' &= \rho S - dV \\
Y' &= \beta_Y SY - dY - pY - \gamma Y \\
T' &= pY - dT - \alpha T - \phi T \\
X' &= \beta_X SX + \alpha T - dX - \delta X - \epsilon X \\
R' &= \gamma Y + \phi T + \epsilon X - dR \\
E_Y' &= \nu Y - \mu E \\
E_X' &= wX - \mu E
\end{aligned}
\tag{1}
$$

Briefly, susceptibles (S) are born into the population with rate λ and they can die with rate d from natural causes. Suscepbles can be infected by people that are infected with drug sensitive (Y) and drug resistant (X) strains. They can also be infected by environmental reservoirs for either strain ((E_Y, E_X). Infected individuals X and Y can die of natural causes, but we assume that those infected with the drug resistant strain can also die with rate δ. Y individuals can also be treated with rate p and move to the treatment class (T). Individuals in the treatment class can die with natural causes, but they can also develop a drug resistant infection with rate α. All infected classes can recover to the recovered class (with rates γ, ϵ, ϕ). Infected individuals can shed bacteria into the environmental reservoir with rates ν and w from the Y and X classes. Recovered (R) individuals can die from natural causes and the bacteria can die with rate μ in the environment. Finally, susceptible can be vaccinated against either bacterial strain. This occurs with rate ρ. Vaccinated individuals can also die from natural causes.

Table 1: Model parameters and variables

	Description	Initial Value	Ref
S	Susceptible Population	N	
V	Vaccinated Population	0	
Y	Infected Population	1	
T	Treatment Population	0	
X	Infected Population with Resistant Strain	0	
R	Recovered Population	0	
E_Y, E_X	Environment reservoirs for the drug sensitive and drug resistant strains	0	
N	Total Population, $N = S + V + Y + X + T + R$	10000	
		Value	
λ	Birth rate	d*N people*day^{-1}	
d	Death rate	1/(82*52*7) day^{-1}	REF6
$\beta_{Y,X}$	Infection rates from humans in the X and Y classes	Varied (people*day)$^{-1}$	
β_e	Infection rate from environmental reservoir	Varied (volume*day)$^{-}$	

		[1]	
ρ	Vaccination rate of S	Varied day^{-1}	
p	Treatment rate of Y	Varied day^{-1}	
γ, ϵ	Recovery rate of Y, X	1/10 day^{-1}	REF7
ϕ	Recovery rate of T	1/5 day^{-1}	REF8
α	Rate of development of a resistant infection from Y	Vary	
ν, w	Shedding rate of pathogen into the environmental reservoir from Y and X	Varied volume*(people*day)$^{-1}$	
μ	Decay rate of pathogen in the environmental reservoir	13.2–106.5 day^{-1}	REF9

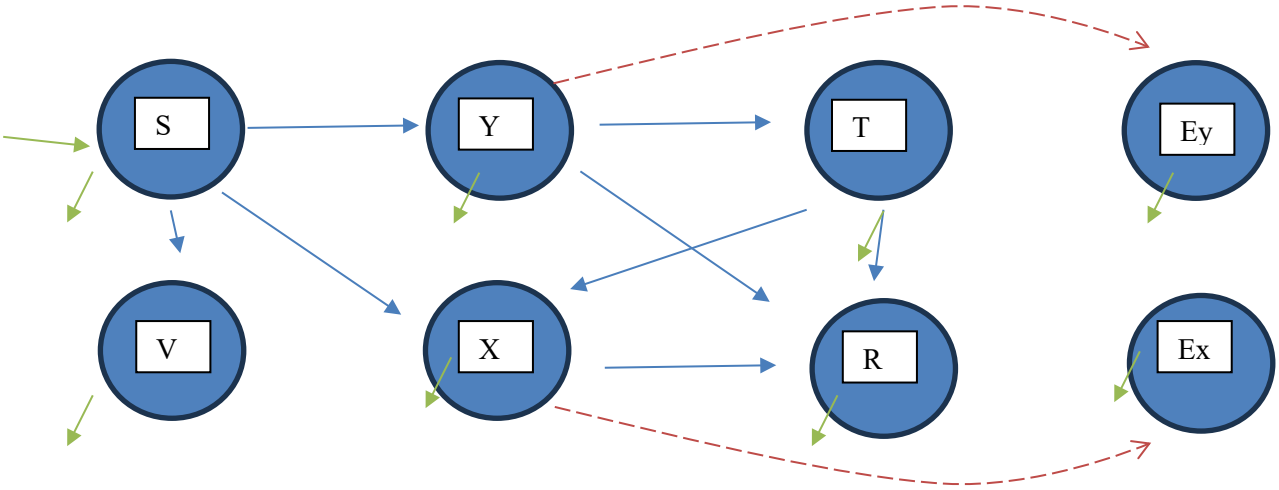


Figure 1: Model diagram. Blue arrows denote change in infected state. Red arrows denote shedding into environmental reservoirs. Green arrows denote birth and death.

4. Preliminary Results

[I chose to include some analysis for my preliminary results. I could have instead chosen to run some simulations and describe their graphs in paragraph form. Present your results in sentence form. If you have figures or tables in this section, you must discuss them in text.]

4.1 A Reduced Model Ignoring the Environmental Reservoir

Consider the submodel of Equation (1) that ignores the environmental reservoirs Ex and Ey. That is, we set Ex and Ey to be zero. Let N represent the number of people in the population. Then N = S + Y + T + V + X + R and

$$\begin{aligned} N' &= S' + Y' + T' + V' + X' + R' \\ &= \lambda - dS - dY - dT - dV - dX - dR - \delta X \\ &= \lambda - dN - \delta X \end{aligned}$$

Ignoring the death due to disease (δ), we find that $\frac{dN}{dt} = 0$ when $\lambda = dN$. That is, the population is constant when the number of births per time (λ) and the number of natural deaths per time (dN) are equal.

The equilibria (or steady states) of the reduced system are found by solving

$$\begin{aligned} 0 &= \lambda - dS - \beta_Y SY - \beta_X SX - \rho S \\ 0 &= \rho S - dV \\ 0 &= \beta_Y SY - dY - pY - \gamma Y \\ 0 &= pY - dT - \alpha T - \phi T \\ 0 &= \beta_X SX + \alpha T - dX - \delta X - \epsilon X \\ 0 &= \gamma Y + \phi T + \epsilon X - dR \end{aligned} \tag{2}$$

Solving this system, we find that we have three equilibria: a disease-free equilibrium

$$Q_0 = \left\{ S = \frac{\lambda}{d+\rho}, Y = 0, T = 0, V = \rho\frac{\lambda}{(d+\rho)d}, X = 0, R = 0, \right\}$$

and two infected equilibria

$$Q_X = \left\{ \begin{array}{c} S = \frac{d+\delta+\epsilon}{\beta_X}, Y = 0, T = 0, V = \frac{\rho(d+\delta+\epsilon)}{\beta_X d}, \\ X = \frac{\beta_X\lambda - d^2 - d\delta - d\epsilon - d\rho - \delta\rho - \epsilon\rho}{\beta_X(d+\delta+\epsilon)}, R = \frac{\epsilon(\beta_X\lambda - d^2 - d\delta - d\epsilon - d\rho - \delta\rho - \epsilon\rho)}{\beta_X(d+\delta+\epsilon)d} \end{array} \right\}$$

where the we only have the drug resistant strain, and

$$Q_{XY} = \left\{ \begin{array}{c} S = \frac{d+\gamma+p}{\beta_Y}, \\ Y = -\frac{(d^2+(p+\rho+\gamma)d - \beta_Y\lambda + \rho(p+\gamma))(\alpha+d+\phi)(\beta_X - \beta_Y)d + (-\epsilon-\delta)\beta_Y + \beta_X(p+\gamma)}{(d+\gamma+p)\left((\beta_X-\beta_Y)d^2 + ((-\epsilon-\delta-\phi-\alpha)\beta_Y + \beta_X(p+\phi+\alpha+\gamma))d - (\phi+\alpha)(\epsilon+\delta)\beta_Y + ((\phi+\alpha)\gamma+p\phi)\beta_X\right)\beta_Y} \\ T = -\frac{(d^2+(p+\rho+\gamma)d - \beta_Y\lambda + \rho(p+\gamma))p(\beta_X - \beta_Y)d + (-\epsilon-\delta)\beta_Y + \beta_X(p+\gamma)}{(d+\gamma+p)\left((\beta_X-\beta_Y)d^2 + ((-\epsilon-\delta-\phi-\alpha)\beta_Y + \beta_X(p+\phi+\alpha+\gamma))d - (\phi+\alpha)(\epsilon+\delta)\beta_Y + ((\phi+\alpha)\gamma+p\phi)\beta_X\right)\beta_Y} \\ V = \rho\frac{d+\gamma+p}{\beta_Y d} \\ X = \frac{(d^2+(p+\rho+\gamma)d - \beta_Y\lambda + \rho(p+\gamma))p\alpha}{(d+\gamma+p)\left((\beta_X-\beta_Y)d^2 + ((-\epsilon-\delta-\phi-\alpha)\beta_Y + \beta_X(p+\phi+\alpha+\gamma))d - (\phi+\alpha)(\epsilon+\delta)\beta_Y + ((\phi+\alpha)\gamma+p\phi)\beta_X\right)\beta_Y} \\ R = -\frac{\begin{array}{c}(d^2+(p+\rho+\gamma)d - \beta_Y\lambda + \rho(p+\gamma))\gamma(\beta_X - \beta_Y)d^2 + \left(((-\epsilon-\delta-\phi-\alpha)\gamma - p\phi)\beta_Y + (\gamma^2 + (p+\phi+\alpha)\gamma + p\phi)\beta_X\right)d \\ + \\ (-(\phi+\alpha)(\epsilon+\delta)\gamma - ((\epsilon+\delta)\phi + \alpha\epsilon)p)\beta_Y + (p+\gamma))((\phi+\alpha)\gamma+p\phi)\beta_X\end{array}}{(d+\gamma+p)\left((\beta_X-\beta_Y)d^2 + ((-\epsilon-\delta-\phi-\alpha)\beta_Y + \beta_X(p+\phi+\alpha+\gamma))d - (\phi+\alpha)(\epsilon+\delta)\beta_Y + ((\phi+\alpha)\gamma+p\phi)\beta_X\right)\beta_Y * d} \end{array} \right\}$$

that includes the drug sensitive and resistant strains. Note that Q_0 is always positive. Q_X is positive when

$$\beta_X\lambda - d^2 - d\delta - d\epsilon - d\rho - \delta\rho - \epsilon\rho > 0$$

There are multiple criteria that must be satisfied for Q_{XY} to be positive. We do not list them here.

The disease free equilibrium Q_0 is desirable as it means that there is no bacteria in the system. We now look to see when this equilibrium is locally stable. The Jacobian of the system is

$$J := \begin{bmatrix} -XbX - YbY - d - \rho & 0 & -SbY & 0 & -bXS & 0 \\ \rho & -d & 0 & 0 & 0 & 0 \\ YbY & 0 & SbY - d - \gamma - p & 0 & 0 & 0 \\ 0 & 0 & p & -\alpha - d - \phi & 0 & 0 \\ XbX & 0 & 0 & \alpha & bXS - d - dd - \epsilon & 0 \\ 0 & 0 & \gamma & \phi & \epsilon & -d \end{bmatrix}$$

Evaluating the Jacobian at Q_0 gives

$$J00 := \begin{bmatrix} -d - \rho & 0 & -\frac{\lambda bY}{d+\rho} & 0 & -\frac{bX\lambda}{d+\rho} & 0 \\ \rho & -d & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\lambda bY}{d+\rho} - d - \gamma - p & 0 & 0 & 0 \\ 0 & 0 & p & -\alpha - d - \phi & 0 & 0 \\ 0 & 0 & 0 & \alpha & \frac{bX\lambda}{d+\rho} - d - dd - \epsilon & 0 \\ 0 & 0 & \gamma & \phi & \epsilon & -d \end{bmatrix}$$

Using this Jacobian, we obtain six eigenvalues:

$$\left\{-(d+\rho), -(\alpha+d+\phi), -d, -d, \frac{\beta_X\lambda - d^2 - d\delta - d\epsilon - d\rho - \delta\rho - \epsilon\rho}{d+\rho}, -\frac{-\beta_Y\lambda + d^2 + d\gamma + dp + d\rho + \gamma\rho + p\rho}{d+\rho}\right\}$$

Note that the first four eigenvalues are always negative since $d, \rho, \alpha, \phi > 0$. Eigenvalues five and six are negative when

$$R_{0,X} = \frac{\beta_X\lambda}{(d+\delta+\epsilon)(d+\rho)} < 1$$
$$R_{0,Y} = \frac{\beta_Y\lambda}{(d+\gamma+p)(d+\rho)} < 1$$

We define $R_{0,X}$ and $R_{Y,0}$ as the basic reproduction numbers for the X and Y strains. Note that $R_{0,X} > 1$ satisfies the condition $\beta_X\lambda - d^2 - d\delta - d\epsilon - d\rho - \delta\rho - \epsilon\rho > 0$, which provides the existence criteria for infected equilibrium Q_X.

5. Future work

My full report will include similar analysis to what is presented in Section 4, but for the full model with the environmental reservoirs. I will also run some numerical simulations for different parameter sets to see how we might be able to control e-coli in the system.

References

[Here you must list the references that you used. Use proper citation formatting.]